

A Method for Blind Detection of OFDM Signal Based on Power Spectrum Reprocessing

ZHANG Hai-ying; YUAN Chao-wei

*Institute of Telecommunication and Network integrated Technology, Beijing University of Posts
and Telecommunications, Beijing 100876, China*

E-mail:ellen_lily@126.com

Abstract

A method for blind detection of OFDM signal based on power spectrum reprocessing is presented for distinguishing OFDM signal and single carrier modulation signals. The method takes full advantage of cryptic period of OFDM signal and enhances its ability for processing through power spectrum pretreatment, and has strong ability to suppress additive interference and is applicable to frequency selectivity fading channel and accurately estimate subcarrier frequency interval, symbol period, signal bandwidth and subcarrier number of OFDM signal. The simulation result shows that the new method is able to detect OFDM signals rapidly and accurately in condition of relatively low signal-noise ratio, and achieves high identification probability, has high adaptability for frequency selectivity fading channel, and has also low computational complexity, high precision, and is suitable for hardware implement.

1. Introduction

As a kind of modulation scheme for resolving high rate data transmission over wireless multi-path channel, OFDM is widely used in civil or military communication. Therefore, the research on realization of OFDM signal blind detection under the condition without any prior knowledge becomes key technical issue to be resolved urgently in the field of information countermeasure.

OFDM adopts multiple subcarriers with equal frequency interval to form parallel transmission system, every channel of subcarrier modulates one channel of separate data information, signals on each adjacent subcarrier overlap on the spectrum, their amplitudes approximately accord with Gaussian distribution. Currently existing methods basically use the feature to distinguish OFDM and other single-carrier signal. Literature [1] use classification features of high order matrix or high order mixed matrix to identify OFDM

signals with Gaussian distribution and other signals with non-Gaussian distribution. The method is only suitable for additive Gaussian noise, can't effectively suppress multiplicative interference caused by fading channel. Based on the feature that OFDM signal multi-scale wavelet decomposition coefficient varies little, literature [2] presents using signal multi-scale wavelet decomposition as characteristic vector. Its shortcoming is requirement of multi-resolution wavelet decomposition and large operation.

Aiming at frequency domain characteristics of OFDM signal, the paper presents a method of power spectrum reprocessing based OFDM signal blind detection and identification. The method fully uses cryptic period of OFDM signal spectrum and adopts evolution function to process the power spectrum in order to suppress the influence of multi-path channel, can rapidly and accurately detect and identify OFDM signal under the condition of relatively low SNR, meanwhile estimate the signal's sub-carrier frequency interval and sub-carrier number, with no prior conditions.

2. Blind Detection Method

2.1 Power spectrum and power spectrum reprocessing of OFDM signal

Complex form of OFDM signal can be expressed as:

$$S_{OFDM}(t) = \frac{1}{N} \sum_{n=1}^N A_n(t) e^{j[\omega_n t + \phi_n(t)]} \quad (1)$$

Where, $\omega_n = \omega_0 + n\Delta\omega$. N is the number of subcarriers, $\Delta\omega$ is subcarrier frequency interval. They are both important parameters to be estimated. $A_n(t)$ and $\phi_n(t)$ are respectively amplitude and phase of each subcarrier. They change with the variety of the symbols, but keep constant during one symbol period

T_c . So, the signals in one symbol are only related to the carrier frequency, i.e.,

$$A_n(t) \Rightarrow A_n, \quad \phi_n(t) \Rightarrow \phi_n \quad (2)$$

Assume $\omega_0 = 0$, the simplified formula of periodical signal of single symbol is:

$$S_s(t) = \frac{1}{N} \sum_{n=1}^N A_n e^{j\phi_n} e^{jn\Delta\omega t} \quad (3)$$

$A_n e^{j\phi_n}$ is the definition of signal in frequency domain, therefore OFDM signal can be expressed with Fourier transform.

When the following equation comes into existence, subcarriers are orthogonal, i.e., integral times of carrier periods are contained in integral time T_c .

$$\Delta f = \frac{\Delta\omega}{2\pi} = \frac{1}{T_c} \quad (4)$$

From equation (3), OFDM signal spectrum in one symbol period-- $F_s(\omega)$ can be obtained:

$$F_s(\omega) = \frac{1}{N} \sum_{n=1}^N A_n \delta(\omega - n\Delta\omega) \quad (5)$$

Its power spectrum $P_s(\omega)$ is:

$$\begin{aligned} P_s(\omega) &= F_s(\omega) F_s^*(\omega) = \left[F_s(\omega) \right]^2 \\ &= \frac{1}{N^2} \sum_{n=1}^N A_n^2 \delta(\omega - n\Delta\omega) \end{aligned} \quad (6)$$

Signal's multiple symbol periods being considered, OFDM signal's spectrum expression with much universality -- $F_{OFDM}(\omega)$ is obtained:

$$\begin{aligned} F_{OFDM}(\omega) &= F_s(\omega) \otimes F_{n,v}(\omega) \\ &= \frac{1}{N} \sum_{n=1}^N A_n \delta(\omega - n\Delta\omega) \otimes F_{n,v}(\omega) \\ &= \frac{1}{N} \sum_{n=1}^N A_n F_{n,v}(\omega - n\Delta\omega) \end{aligned} \quad (7)$$

Its power spectrum $P_{OFDM}(\omega)$ is:

$$\begin{aligned} P_{OFDM}(\omega) &= F_{OFDM}(\omega) \cdot F_{OFDM}^*(\omega) \\ &= \frac{1}{N^2} \sum_{n=1}^N A_n^2 \left| F_{n,v}(\omega - n\Delta\omega) \right|^2 \end{aligned} \quad (8)$$

Where, $F_{n,v}(\omega)$ is Fourier transform of multiple data symbols of n-th subcarrier, can be regarded as weighting factor of n-th subcarrier of $F_s(\omega)$.

$F_{n,v}(\omega)$ varies with data variety. Therefore, the power spectrum of OFDM signal in multiple symbol periods has obvious quasi-periodicity ($\Delta\omega$ is taken as the period).

The autocorrelation function of OFDM signal can be obtained by inverse Fourier transform of equation (8) power spectrum:

$$\begin{aligned} R_{OFDM}(\tau) &= IFFT \left[P_{OFDM}(\omega) \right] \\ &= \frac{1}{N^2} \sum_{n=1}^N A_n^2 \left[IFFT \left(\left| F_{n,v}(\omega - n\Delta\omega) \right|^2 \right) \right] \end{aligned} \quad (9)$$

Power spectrum reprocessing is defined as power spectrum of power spectrum of signal. The technique fully uses quasi-period characteristics in signal frequency domain, tests signal components submerged in the noise, realizes signal detection and parameter estimation, i.e.,

$$\begin{aligned} s(t') &= \left| FFT \left[\left| FFT[S_{OFDM}(t)] \right|^2 \right] \right|^2 \\ &= \left| FFT[P_{OFDM}(\omega)] \right|^2 \end{aligned} \quad (10)$$

Then,

$$S_{OFDM}(\tau) = K \cdot \left| R_{OFDM}(-\tau) \right|^2 = K \cdot \left| R_{OFDM}(\tau) \right|^2 \quad (11)$$

Where, K is a constant, $R_{OFDM}(\tau)$ is even symmetry.

So, power spectrum reprocessing of OFDM signal represents modulus square of autocorrelation function and has higher periodic characteristics than autocorrelation. Spectrum reprocessing can add the nonlinear operation to the power spectrum (such as logarithm, evolution etc.) to suppress multiplicative interference and strengthen the characteristic of spectrum line, therefore can extract signal feature parameters more accurately.

2.2 OFDM signal detection and parameter estimation algorithm flow

The algorithm flow can be reduced to:

① Use Welch modified average periodogram algorithm to estimate the average power spectrum, its length is L ;

② Integrate the average power spectrum, calculate signal bandwidth's two endpoint frequency position number, N_H and N_L , then the signal bandwidth is:

$$B_s = (N_H - N_L) \cdot \Delta f,$$

Where Δf is spectrum resolution, $\Delta f = f_s / L$;

③ Perform Fourier transform to the power spectrum and take modulus square, thus acquire “spectrum reprocessing”. Perform periodicity analysis to spectrum line of spectrum reprocessing, detect whether OFDM signal exists;

④ Measure period value of “spectrum reprocessing” envelope periodic spectrum peak B , M , acquire OFDM signal subcarrier frequency interval:

$$\Delta f_{OFDM} = f_s / M ;$$

⑤ According to signal bandwidth and subchannel frequency interval, acquire the number of subcarriers $\hat{N} = \text{round}(B_s / \Delta f_{OFDM})$.

3 Feature analysis

3.1 Additive noise influence

Assume additive noise signal is

$$x(t) = S_{OFDM}(t) + N(t) \quad (12)$$

Where $N(t)$ is noise, power spectrum is

$P_n(\omega) = N_0$, and the signal is independent of the noise. Thus,

$$P_x(\omega) = P_{OFDM}(\omega) + P_n(\omega) \quad (13)$$

The power spectrum reprocessing is:

$$\begin{aligned} s_x(t') &= |FFT[P_x(\omega)]|^2 \\ &= |FFT[P_{OFDM}(\omega) + N_0]|^2 \\ &= |FFT[N_0] + FFT[P_{OFDM}(\omega)]|^2 \\ &= |FFT[N_0]|^2 + |FFT[P_{OFDM}(\omega)]|^2 \\ &\quad + 2|FFT[N_0]FFT[P_{OFDM}(\omega)]| \\ &= |N_0 \cdot 2\pi \cdot \delta(t')|^2 + K |R_{OFDM}(t')|^2 \\ &\quad + |2N_0 \cdot 2\pi \cdot R_{OFDM}(0) \cdot \delta(t')| \end{aligned} \quad (14)$$

From equation (14), it is known that signal additive noise “spectrum reprocessing” result is related to autocorrelation function of signal and average power of noise. For OFDM signal, its autocorrelation function contains narrow pulse periodically appearing. Narrow pulse is Sa function compressed from OFDM signal with T_c width and its width is about $1/B$, it has $T_c B$ times of gain. But the noise after spectrum reprocessing hasn't processing gain except for zero-point. Therefore, spectrum reprocessing based detecting method improves output SNR, has higher detecting capability.

3.2 Multiplicative noise influence

3.2.1 The model of multi-path channel

Multi-path effect existing in mobile channel will cause signal's selective fading which is a kind of multiplicative interference. Assume the receive signal is sum of multiple signals with various path, i.e.,

$$S_r(t) = \sum_{i=1}^M a_i S_{OFDM}(t - \tau_i(t)) \quad (15)$$

Where, a_i is attenuation coefficient of i-th path;

$\tau_i(t)$ is relative transmission delay of i-th path.

Without loss of generality, take dual-channel as an example, and take the first path as the reference.

Assume signal on path “1” is $S_i(t)$, signal on path

“2” is $rS_i(t + \Delta t)e^{j\omega\Delta t}$, r expresses relative

attenuation coefficient between paths, Δt expresses relative delay between paths. Because Δt is so small

that $S_i(t + \Delta t)$ is approximate to $S_i(t)$, then the

receive signal can be expressed as:

$$S_0(t) = S_i(t)(1 + re^{j\omega\Delta t}) \quad (16)$$

Channel transfer function is:

$$H(t, j\omega) = \frac{S_0(t)}{S_i(t)} = 1 + re^{j\omega\Delta(t)} \quad (17)$$

Channel amplitude feature is:

$$\begin{aligned} A(t, \omega) &= |H(t, j\omega)| \\ &= \sqrt{1 + r^2 + 2r \cos[\omega\Delta(t)]} \end{aligned} \quad (18)$$

3.2.2 Analysis of multi-path processing capability of “spectrum reprocessing”

The signal transmitted on the channel can be expressed as convolution of original signal and channel impulse response, i.e.,

$$\begin{aligned} x(t) &= S_{OFDM}(t) * h(t) \\ &= \sum_{\tau=-\infty}^{+\infty} S_{OFDM}(t)h(t - \tau) \end{aligned} \quad (19)$$

Perform spectrum reprocessing to $x(t)$, make the convolution relationship in time domain convert to multiply relationship in frequency domain, then convert to convolution relationship in time domain, i.e.,

$$\begin{aligned}
s_x(t') &= \left| \text{FFT} \left[\left| F_{\text{OFDM}}(\omega) \right|^2 \left| H(t, j\omega) \right|^2 \right] \right|^2 \\
&= \left| \text{FFT} \left\{ \left| F_{\text{OFDM}}(\omega) \right|^2 \left[(1+r^2) + 2r \cos(\omega\Delta t) \right] \right\} \right|^2
\end{aligned} \quad (20)$$

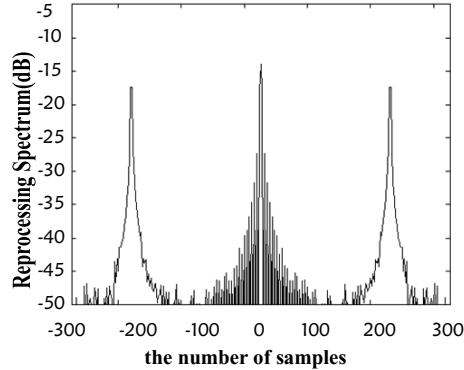
Where,

$$\begin{aligned}
&\text{FFT} \left[(1+r^2) + 2r \cos(\omega\Delta t) \right] \\
&= 2\pi \left\{ (1+r^2) + r \left[\delta(t-\Delta t) + \delta(t+\Delta t) \right] \right\}
\end{aligned} \quad (21)$$

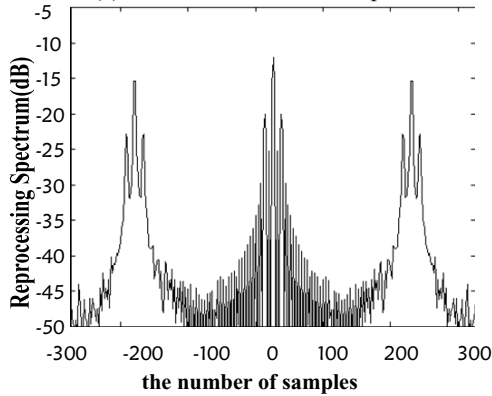
For simplifying the formula, 2π is figured in the item of constant K , then

$$\begin{aligned}
s_x(t') &= (1+r^2)^2 K \left| R_{\text{OFDM}}(\tau) \right|^2 \\
&+ r^2 K \left[\left| R_{\text{OFDM}}(\tau - \Delta t) \right|^2 + \left| R_{\text{OFDM}}(\tau + \Delta t) \right|^2 \right]
\end{aligned} \quad (22)$$

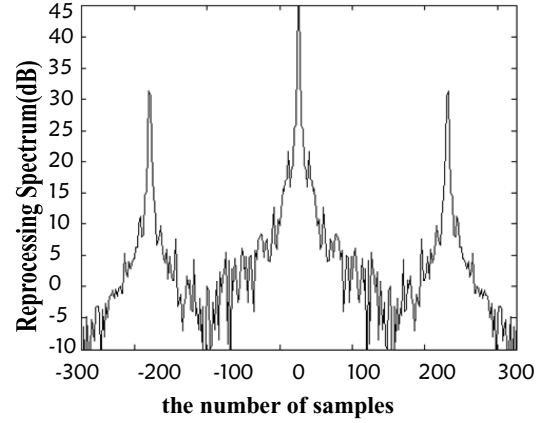
From equation (22), it's known that multi-path channel response can be taken into the autocorrelation function by convolution processing in time domain, then its influence can make the pulse become wide, shown as figure 1(a) and 1(b).



(a) Condition without multi-path



(b) Condition with multi-path



(c) Tested performances of improved spectrum reprocessing method

Fig.1 Influence of fading channel on spectrum reprocessing detection

3.2.3 Improved method for power spectrum reprocessing

In order to repress multi-path influence, homomorphic processing such as cepstrum processing and processing function with compression effect can be added to power spectrum. Evolution has the characteristics of simplification, and is easy to be realized by digital signal processing and overcome the shortcomings of the presented conventional cepstrum algorithms.

The paper adopts processing method of evolution to reduce multi-path influence. Take extraction of square root as an example, conclude as follows:

$$\begin{aligned}
c_x(t') &= \left| \text{FFT} \left[\left| F_{\text{OFDM}}(\omega) \right| \left| H(t, j\omega) \right| \right] \right|^2 \\
&= \left| \text{FFT} \left[\left| F_{\text{OFDM}}(\omega) \right| \sqrt{1+r^2+2r \cos(\omega\Delta t)} \right] \right|^2 \\
&= \left| \text{FFT} \left[\sqrt{P_{\text{OFDM}}(\omega)} \right] \otimes \text{FFT} \left[\sqrt{1+r^2+2r \cos(\omega\Delta t)} \right] \right|^2
\end{aligned} \quad (23)$$

Maclaurin expansion of $(1+x)^m$ function is:

$$\begin{aligned}
(1+x)^m &= 1 + mx + \frac{m(m-1)}{2!} x^2 + \\
&\dots + \frac{m(m-1) \cdots (m-n+1)}{n!} x^n + \dots
\end{aligned} \quad (24)$$

Then, the expansion of extraction of square root is:

$$(1+x)^{\frac{1}{2}} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4 + \dots \quad (25)$$

Expend $\sqrt{1+r^2+2r\cos(\omega\Delta t)}$ to the series and neglect the high-order item, obtain:

$$\begin{aligned} & \left\{1 + \left[r^2 + 2r\cos(\omega\Delta t)\right]\right\}^2 \\ & \approx 1 + \frac{1}{2}\left[r^2 + 2r\cos(\omega\Delta t)\right] - \frac{1}{8}\left[r^2 + 2r\cos(\omega\Delta t)\right]^2 \\ & \approx 1 + \frac{r^2}{4} - \frac{r^4}{8} + r\left(1 - \frac{r^2}{2}\right)\cos(\omega\Delta t) - \frac{r^2}{4}\cos(2\omega\Delta t) \end{aligned} \quad (26)$$

So the improved spectrum reprocessing for multi-path channel response can be expressed as:

$$\begin{aligned} & FFT\left\{1 + \left[r^2 + 2r\cos(\omega\Delta t)\right]\right\}^2 \\ & \approx 2\pi\left\{\left(1 + \frac{r^2}{4} - \frac{r^4}{8}\right)\delta(t) + \frac{r}{2}\left(1 - \frac{r^2}{2}\right)\left[\delta(t - \Delta t) + \delta(t + \Delta t)\right] \right. \\ & \quad \left. - \frac{r^2}{8}\left[\delta(t - 2\Delta t) + \delta(t + 2\Delta t)\right]\right\} \end{aligned} \quad (27)$$

Compare equation (21) with equation (27) (the ratio of the amplitude value of the first order time delay Δt before the test method is improved to that after the test method is improved is $\frac{2}{(1-\frac{r^2}{2})}$, where $|r| \leq 1$).

It shows that multi-path channel influence is reduced 6 dB at least. From equation (24), it's known that the suppression effect would be more obvious if extraction of the fourth root is chosen.

Such being the case, this paper puts forward to carry on improvement method on spectrum-reprocessing that use analytical signals to compute the instantaneous magnitude.

The instantaneous amplitude of signal can be used to calculate power spectrum for extracting envelope information of spectrum reprocessing. The instantaneous amplitude of OFDM signal, $|S_{OFDM}(t)|$, is:

$$\begin{aligned} |S_{OFDM}(t)| &= \frac{1}{N} \left| e^{j\omega_0 t} \left| \sum_{n=1}^N A_n(t) e^{j[n\Delta\omega t + \phi_n(t)]} \right| \right| \\ &= \frac{1}{N} \left| \sum_{n=1}^N A_n(t) e^{j[n\Delta\omega t + \phi_n(t)]} \right| \end{aligned} \quad (28)$$

From the above equation, it's shown that frequency modulation influence can be removed by modulus operation for analytic signal. Fig. 1(c) shows processing result of extracting the fourth root for the power spectrum. Fig. 1(c) being compared with Fig.

1(b), it's obvious that the improved method for spectrum reprocessing can suppress multi-path interference well, and improve detection performance.

4 Computer simulation

4.1 Detection of OFDM signal in Gaussian noise

Experimental conditions: signal-noise ratio is $SNR = -3dB$, the number of subcarriers of OFDM signal is $N = 51$, then signal spectrum reprocessing are shown as Fig. 2.

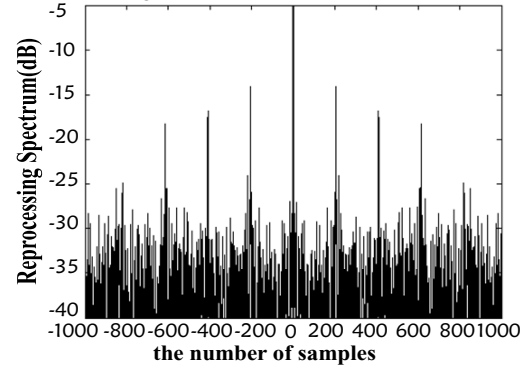
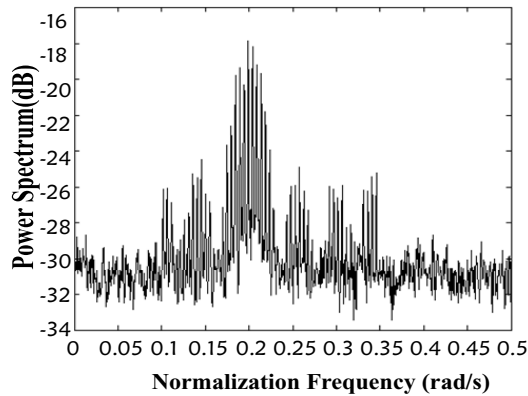


Fig. 2 Additive noise OFDM signal spectrum reprocessing detection result

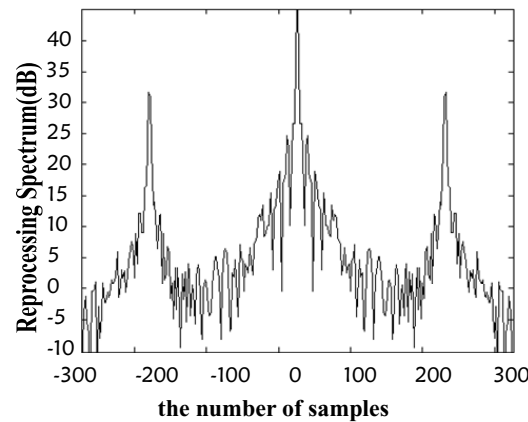
Whether signals exist can't be judged from the power spectrum, but periodic spectrum line can be detected by spectrum reprocessing calculation, so OFDM signal can be judged. While $SNR = -3dB$, the existence of signals can be detected, and subcarrier frequency interval can be estimated.

4.2 Influence on detection performance in fading channel

Main factors of multiplicative interference include: maximum multi-path time delay τ_{max} , number of multi-path channels M , maximum fading depth A_{dmax} . Experimental conditions: $SNR = 10dB$, $\tau_{max} = 0.4 * \tau_{guard}$, $M = 10$, $A_{dmax} = 15dB$, then the signal power spectrum and modified spectrum reprocessing are shown as Fig.3.



(a) Power spectrum



(b) Detection result of modified spectrum reprocessing
Fig. 3 Detection result of OFDM signal on fading channel

When $\tau_{\max} < \tau_{\text{guard}}$, multi-path channel response in the improved spectrum reprocessing is mostly suppressed by adopting processing function with compression effect —extraction of high power root of the power spectrum, so OFDM signal response isn't affected by channel fading basically, and already has whole periodicity. It indicates that the improved spectrum reprocessing detection method has strong capability of suppressing multi-path interference.

5 Conclusion

Power spectrum reprocessing based OFDM signal blind detection method presented in the paper modifies spectrum reprocessing method by adopting instantaneous amplitude and performing evolution to power spectrum, not only strips the frequency modulation influence by using analytic signals and

suppresses multi-path interference by extraction of high power root of power spectrum, but also suppresses signals with no cryptic period such as single-carrier modulation single and noise, etc. by periodic detection for spectrum reprocessing. Therefore it can perform detection and parameter estimation for OFDM signal at low SNR. Theoretical analysis and simulation experiment all prove the availability of the method.

6 Reference

- [1] Wang bin, Ge lindong. An algorithm of blind detection of high order matrix based OFDM signal modulation[J], Data Collection and Processing, 2006, 21(1): 37-41.
- [2] Han gang, Research on technique of signal blind detection in selfadaptive single-carrier multi- carrier modulation[D], Xi'an: Xidian University, Doctoral Dissertation, 2003.
- [3] Mucchi L.; Morosi, S.; Del Re, E.; Fantacci, R. A new algorithm for blind adaptive multiuser detection in frequency selective multipath fading channel[J]. IEEE Transactions on Wireless Communications, 2004, 3(1):235-247.
- [4] Del Re, E.; Fantacci, R.; Morosi, S.; Pugi, A. Advanced blind adaptive multi-user detector for communications in nonstationary multipath fading channel [J], IEEE Transactions on Vehicular Technology ,2001, 50(6): 1497-1506.
- [5] Chang Tianqi; Zhou Zhengzhong. A spectral method for periodic estimation of the PN sequences in lower SNR DS-CDMA signals[C]. IEEE 2002 International Conference on Communications, Circuits and Systems and West Sino Expositions, Chengdu, China, IEEE, 2002: 506-510 .
- [6] Hess W. Pitch Determination of Speech Signals [M]. New York: Springer-Verlag, 1983.
- [7] Akmouche W. Detection of multicarrier modulations using 4th order cumulants [C]. Proc of IEEE MILCOM'99. Atlantic City, NJ, USA: IEEE, 1999: 432-436.
- [8] Giannakis G B, Tsatsanis M K. Time-domain tests for gaussianity and time reversibility [J]. IEEE Trans on Signal Processing, 1994, 42 (11): 3460- 3472.